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Department of Computer Science
COMSATS University Islamabad, Lahore Campus

Midterm Examination– Semester Spring 2025

Course Title:	Computer Networks			Course Code:	CSC275	Credit Hours:	3(2,1)
Course Instructor/s:	Mr. Imran Raza, Dr. Hasanain Ch., Ms. Hina Saleem, Ms. Sidra Nasir, Mr. Naseer Akhlaq, Mr. Qasim Ali			Program Name:	BS (CS) & BS (SE)		
Semester;	4th	Batch: FA23	Section:	All Sections	Date:	15-04-2025	
Time Allowed:				Maximum Marks:	25		
Name:				Registra'			
Instructions:							

1. Answer any three o… …t questions. [*CLO:1;* Summarize the fundamental concepts of computer networks*; Bloom Taxonomy Level:<Understanding>*] [15]

a. What difficulties are faced in streaming stored video content, and what potential remedies exist? How does Dynamic Adaptive Streaming over HTTP (DASH) address these challenges? (5)

b. Explain the differences between Transport Layer Security (TLS) and Secure Sockets Layer (SSL) in terms of their functionality, using examples to illustrate your explanation. Additionally, specify which version of TLS/SSL you would prefer and justify your choice. (5)

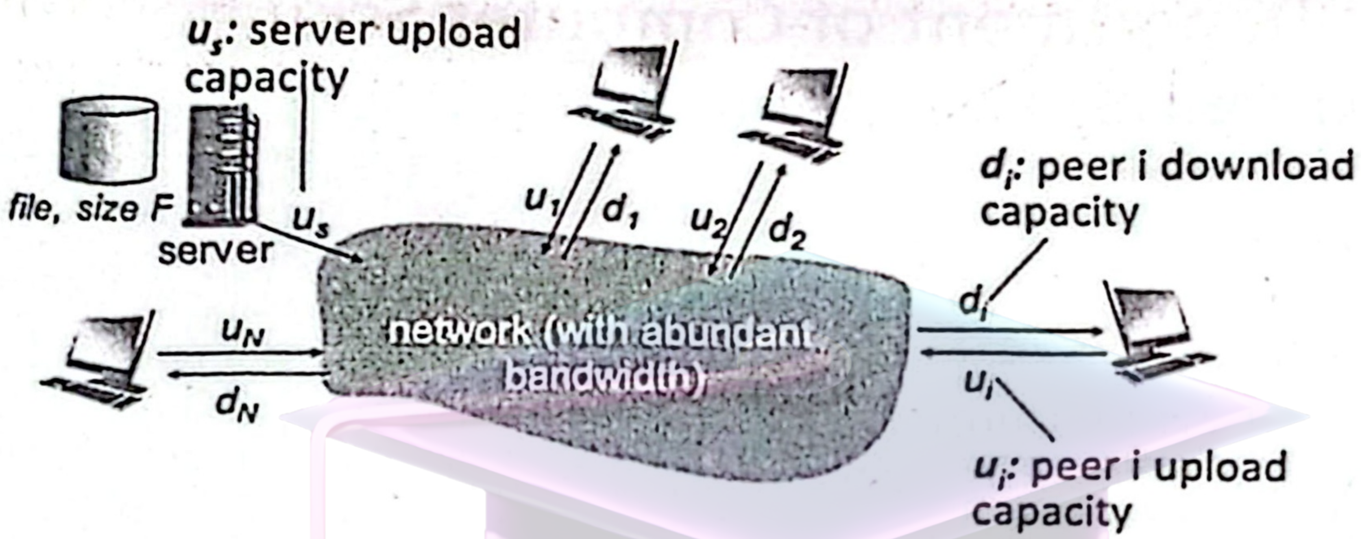
c. Compare HTTP/1.1, HTTP/2, and HTTP/3 in terms of their main features and performance improvements. Discuss aspects such as connection management, latency optimization, and changes in the transport protocol. Additionally, explain key concepts like binary framing and Head-of-Line (HoL) blocking, etc. (5)

d. What server placement philosophies and selection strategies are used in CDNs, and which are preferred by major streaming services like Netflix, YouTube, and Amazon Prime Video? (5)

e. Outline the security challenges faced in DNS queries, including both recursive and iterative approaches, and suggest potential solutions to mitigate these issues? (5)

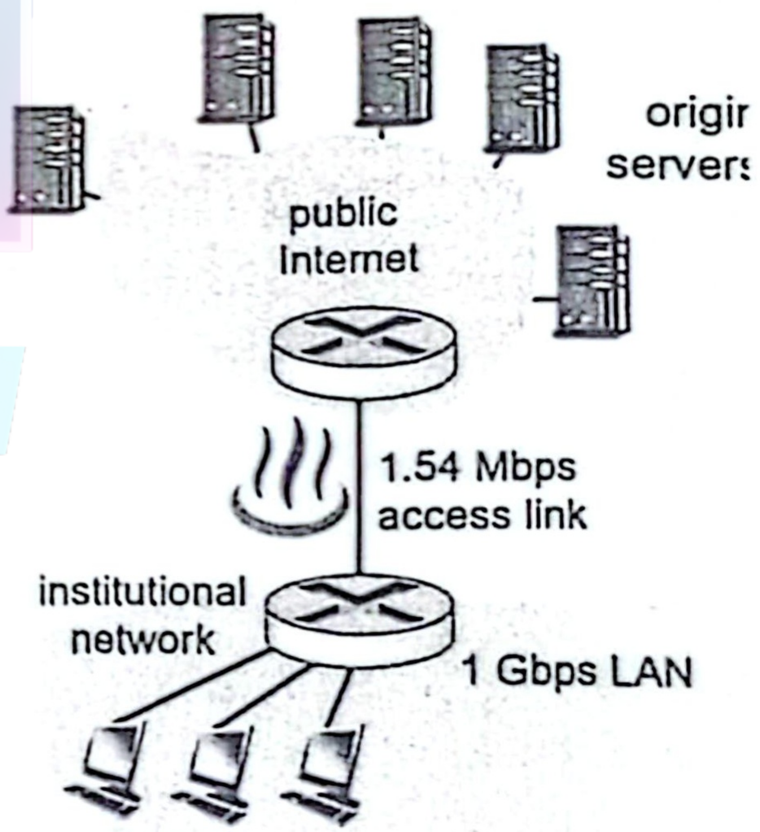
2. Apply your understanding of file distribution to compare client-server and peer-to-peer file-sharing applications. Evaluate the scalability of these applications by considering factors such as file size (F), server's sending rate (u_s), client's download rate (d_i), client's upload rate (u_i), and the number of peers (N). Make reasonable assumptions as needed to support your argument. Additionally, determine how many servers needed to match the file distribution time of a peer-to-peer system and propose a generalized formula for calculating the required number of servers. [CLO:2; Apply the concepts of computer networks to solve its related problems; Bloom Taxonomy Level:<Applying>] [5]

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3. **Analyse the provided scenario to calculate the access link utilization and end-to-end delay. Discuss how implementing a web cache can improve traffic intensity/link utilization and reduce total delay. Assume a cache hit rate of 0.3. [CLO:3; Analyse various protocols, algorithms, and their functions in a network. :<Analysing>]** **[5]**

- ***Access link rate: 1.54 Mbps***
- ***RTT from institutional router to server: 2 seconds***
- ***Web object size: 100K bits***
- ***Average request rate from browsers to origin servers: 15 requests/second***
- ***Average data rate to browsers: 1.50 Mbps***



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